



VCS VERIFICATION REPORT

HEILONGJIANG WANGYUNFENG WIND POWER PROJECT

MONITORING PERIOD:
20 JANUARY 2010 TO 30 JANUARY 2012

REPORT No. 2013-9027
REVISION No. 01

Project Title	Heilongjiang Wangyunfeng Wind Power Project
Version	01
Report ID	Report No. 2013-9027

Report Title	VCS Verification Report for Heilongjiang Wangyunfeng Wind Power Project
DNV Ref.	PRJC-416311-2012-CCS-CHN
Client	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.
Pages	29
Date of Issue	23 April 2013
Prepared By	DNV Climate Change Services AS
Contact	Veritasveien 1, 1322 HØVIK, Norway Tel: +47 67 57 99 00 Email: climatechange@dnv.com http://www.dnv.com
Approved By	Michael Lehmann
Work Carried Out By	YANG Xiaoshan
Technical review by:	Li Lei

Summary:

DNV Climate Change Services AS (DNV) has performed the verification of the emission reductions reported for the project activity “Heilongjiang Wangyunfeng Wind Power Project” for the period 20 January 2010 to 30 January 2012, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. The project is registered by the CDM Executive Board under CDM Ref 5711 and the current monitoring period covers the period pre-CDM registration. The project activity was validated as CDM project by Bureau Veritas Certification Holding SAS (BVCH) (validation Report No.: CHINA-Val/0231/2009, revision 01 dated 4 January 2012) and it was registered as a CDM project on 31 January 2012 under the UNFCCC reference number 5711 with a first 7-year CDM renewable crediting period from 31 January 2012 to 30 January 2019. As emission reductions occur prior to the registration of the project as the CDM project activity, the emission reductions prior to the project's CDM registration cannot be claimed as Certified Emission Reductions (CERs). These emission reductions are thus claimed as Verified Carbon Units (VCU) under the Verified Carbon Standard (VCS) version 3.3. This is in accordance to the eminent VCS Guidance for projects that are registered in two GHG programs.

In addition, DNV has also carried out a validation of the project with regard to VCS requirements that apply to projects already registered under the CDM. DNV was able to confirm that the project is in compliance with the VCS requirements for validation of projects already registered under the CDM.

The verification was performed on the basis of VCS Programme Guide version 3.4 & VCS Standard version 3.3 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspection, and the resolution of outstanding issues.

In our opinion, the GHG emission reductions reported for the project in the monitoring report (version 02 dated 3 April 2013) are fairly stated. The GHG emission reductions were calculated correctly on the basis of approved methodology ACM0002 (version 12.1.0) and the monitoring plan contained in the registered CDM-PDD of 1 August 2011.

Hence, DNV Climate Change Services AS (DNV) is able to certify that the emission reductions from the “Heilongjiang Wangyunfeng Wind Power Project” during the period amount to 195 043 tonnes CO₂ equivalent.

DNV does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to DNV and the engagement conditions detailed in this report. DNV cannot be held liable by any party for decisions made or not made based on this report.

Table of Contents

1	INTRODUCTION	1
1.1	OBJECTIVE	1
1.2	SCOPE AND CRITERIA	1
1.3	LEVEL OF ASSURANCE	2
1.4	SUMMARY DESCRIPTION OF THE PROJECT	2
2	VALIDATION PROCESS, FINDINGS AND CONCLUSION	3
2.1	VALIDATION PROCESS	3
2.1.1	METHOD AND CRITERIA	3
2.1.2	DOCUMENT REVIEW	3
2.1.3	INTERVIEWS	3
2.1.4	SITE INSPECTIONS	3
2.1.5	RESOLUTION OF ANY MATERIAL DISCREPANCY	3
2.2	VALIDATION FINDINGS	4
2.2.1	GAP VALIDATION	4
2.2.2	METHODOLOGY DEVIATIONS	6
2.2.3	PROJECT DESCRIPTION DEVIATIONS	6
2.2.4	NEW PROJECT ACTIVITY INSTANCES	6
2.3	VALIDATION CONCLUSION	6
3	VERIFICATION PROCESS	6
3.1	METHOD AND CRITERIA	6
3.2	DOCUMENT REVIEW	7
3.3	INTERVIEWS	8
3.4	SITE INSPECTIONS	8
3.5	RESOLUTION OF ANY MATERIAL DISCREPANCY	8
4	VERIFICATION FINDINGS	9
4.1	PROJECT IMPLEMENTATION STATUS	9
4.2	ACCURACY OF GHG EMISSION REDUCTION OR REMOVAL CALCULATIONS	14
4.3	QUALITY OF EVIDENCE TO DETERMINE GHG EMISSION REDUCTIONS OR REMOVALS	14
4.4	MANAGEMENT AND OPERATIONAL SYSTEM	16
5	VERIFICATION CONCLUSION	17
6	REFERENCES	19
APPENDIX A		CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

1 INTRODUCTION

Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd. (which is the consultant of the project signed by the project owner /2/ /3/) has commissioned DNV Climate Change Services AS (DNV) to carry out the verification and certification of emission reductions reported for the “Heilongjiang Wangyunfeng Wind Power Project” (the project) in the period 20 January 2010 to 30 January 2012. This report contains the findings from the validation and verification and includes a validation statement for the project and verification statement for the verified carbon units.

1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the “Heilongjiang Wangyunfeng Wind Power Project” for the period 20 January 2010 to 30 January 2012.

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the VCS project document (VCS PD). The VCS PD is reviewed against the criteria stated in the VCS Version 3.3 and the relevant documents and policy announcements made by the VCSA. The Heilongjiang Wangyunfeng Wind Power Project is registered as a CDM project activity (UNFCCC Reference Number: 5711). Therefore, DNV only performed validation of VCS requirement which were not already validated as part of the CDM validation of the project.

The scope of the verification is:

- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan.
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement.
- To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

- VCS Program Guide version 3.4 /26/
- VCS Standard version 3.3 /25/ and other relevant requirements defined by VCSA.
- The approved methodology ACM0002 (version 12.1.0). /23/

The verification shall ensure that reported emission reductions are complete and accurate in order to be verified /22/.

1.3 Level of assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. DNV applied a materiality threshold of 5%/1% with respect to omission or misstatements concerning reported quantities as per VCS standard.

1.4 Summary Description of the Project

1.4.1 Project Category

According to the VCS Program Guide version 3.4 /26/ and the list of Sectoral Scopes of the UNFCCC, the project is applicable under the following activity categories:

- Category 1 – Renewable energy (wind, PV, solar, thermal, biomass, liquid biofuels, geothermal, run-of-river hydro).

According to Annex A of the Kyoto Protocol, the project is applicable under the sectoral scope 1

- Energy Industries (renewable/ non-renewable sources).

1.4.2 Geographic Location

The project is located in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province of China as per the registered CDM-PDD /2/.

1.4.3 Project Background

Title of the project activity:	Heilongjiang Wangyunfeng Wind Power Project
Project Participants:	Hegang Longyuan Wind Power Co., Ltd. authorized by Host Party China
Location of the project activity:	Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province of China
VCU Verification period:	20 January 2010 to 30 January 2012
VCS Project Crediting period:	20 January 2010 to 30 January 2012
UNFCCC Refer. No:	5711
CDM Baseline and monitoring	
Methodology:	ACM0002 (version 12.1.0)
CDM Crediting period:	31 January 2012 to 30 January 2019 (Renewable)
CDM Project Starting date:	12 May 2009 (when the first financial commitment contract was signed) /2/ /3/ /11/
Project Start date:	20 January 2010 /10/ /17/

2 VALIDATION PROCESS, FINDINGS AND CONCLUSION

Heilongjiang Wangyunfeng Wind Power Project has been registered as a CDM project activity (UNFCCC Reference Number: 5711). Therefore, DNV only performed validation for the project on those additional requirements by VCSA. This validation was completed as part of the current VCU verification.

2.1 Validation Process

2.1.1 Method and Criteria

The validation consisted of the following three phases:

- I a desk review of the project design documents
- II follow-up interviews with project stakeholders
- III the resolution of outstanding issues and the issuance of the validation conclusion

The following sections outline each step in more detail.

The criteria of the validation are:

- VCS Program Guide version 3.4 /26/
- VCS Standard version 3.3 /25/ and other relevant requirements defined by VCSA.
- The approved methodology ACM0002 (version 12.1.0) /23/

The validation shall ensure that reported emission reductions are generated by a project that has been able to demonstrate its compliance with the VCSA requirements.

2.1.2 Document Review

The documentation was reviewed in combination with the verification activity for the project. Refer to details in Section 3.2.

2.1.3 Interviews

The interviews were performed in combination with the verification activity for the project. Refer to details in Section 3.3.

2.1.4 Site Inspections

The site inspection was performed in combination with the verification activity for the project. Refer to details in Section 3.4.

2.1.5 Resolution of Any Material Discrepancy

This activity was performed in combination with the verification activity for the project. Refer to details in Section 3.5.

2.2 Validation Findings

2.2.1 Gap Validation

As the project has been validated under the CDM, only the cover page and sections 1.2, 1.3, 1.5, 1.6, 1.7, 1.9, 1.10, 1.12.1, 1.12.2, 1.12.3, 1.12.4 and 1.13 of the *VCS Project Description Template* shall be completed. DNV has undertaken a validation of same, as detailed below.

Cover page

DNV verified relevant information provided in the registered CDM-PDD.

Project title: Heilongjiang Wangyunfeng Wind Power Project

Version: version 02

Date of issue: 1 August 2011

Prepared By: Li Gang, Longyuan (Beijing) Carbon Asset Management Technology Co. Ltd..

Contact: Floor 7, Tower C, International Investment Building, No. 6-9 Fuchengmen North Street, Xicheng District, Beijing 100034, China; telephone +8610-66091326; Ligang@clypg.com.cn

1.2: Sectoral Scope and Project Type

According to the "VCS Sectoral Scopes" /27/, the project is applicable under the sectoral scope 1, Energy (renewable). The project is a grid-connected wind farm power plant with the installed capacity of 49.3 MW.

1.3: Project Proponent

Project Participants: Hegang Longyuan Wind Power Co., Ltd.

1.5: Project Start Date

The project started generating electricity on 20 January 2010 /10/ /17/, therefore, the project start date for VCU issuance is identified as 20 January 2010. The CDM validation of the project was concluded on 4 January 2012, i.e. date of the CDM validation report by Bureau Veritas Certification Holding SAS (BVCH) /3/. Validation of the project was thus completed within two years of the project start date.

1.6: Project Crediting Period

The project crediting period for VCU issuance is determined as from 20 January 2010 to 30 January 2012.

1.7: Project Scale and Estimated GHG Emission Reductions or Removals

Project	Yes
Large project	NA

Years	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 2010 (20/01/2010-31/12/2010, i.e. 346 days)	117 005
Year 2011 (01/01/2011-31/12/2011, i.e. 365 days)	123 430
Year 2012 (01/01/2012-30/01/2012, i.e. 30 days)	10 145
Total estimated ERs (in 741 days)	250 580

Annual 123 430 tCO₂e was estimated in the registered CDM-PDD /2/, from which figures in table above were calculated.

1.9: Project Location

Location of the project activity: Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province of China .

1.10: Conditions Prior to Project Initiation

The project activity is the installation of a new grid-connected renewable wind power plant. The conditions prior to project initiation is that the equivalent electricity delivered to the grid by the project activity which would have otherwise been generated by the operation of grid *connected power plants in the* NECPG.

1.12.1: Proof of Title

Evidences of proof of title have been verified via documentation proving ownership of the wind power plant and equipment at the on-site visit, such as business license of the project owner /14/ agreements signed by the project owner /6/ /11/ and the letter of approval issued by NDRC (DNA of China) /8/.

1.12.2: Emissions Trading Programs and Other Binding Limits

The project is not included in an emissions trading program, and does not take place in a jurisdiction or sector in which binding limits are established on GHG emissions. The project does not reduce GHG emissions from activities that participate in an emissions trading program, so this clause is not applicable.

1.12.3: Participation under Other GHG Programs

The project has been registered as a CDM project on 31 January 2012 under the UNFCCC reference number 5711. The project activity was validated by Bureau Veritas Certification Holding SAS (BVCH) (validation Report No.: CHINA-Val/0231/2009, revision 01 dated 4 January 2012) /3/.

1.12.4: Other Forms of Environmental Credit

The project was registered as a CDM project activity on 31 January 2012 under the UNFCCC reference number 5711. The project has not been registered under any other GHG programs. The project neither has nor intends to generate any other form of GHG related environmental credits for GHG emission reductions or removals, 195 043 tCO₂e, achieved during the period from 20 January 2010 to 30 January 2012 claimed under the VCS Program /1/.

1.13: Additional Information Relevant to the Project

Eligibility Criteria:

The project is not a grouped project, so this clause is not applicable.

Leakage Management:

As per ACM0002 (version 12.1.0), no leakage emissions are considered. The main emissions potentially giving rise to leakage in the context of electric sector projects are emissions arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, and transport). These emissions sources are neglected. Hence this clause is not applicable.

Commercially Sensitive Information

It has been confirmed that there is no any commercially sensitive information excluded from the public version of the project description. Hence this clause is not applicable.

Further Information

As a wind energy project, the project activity will produce clean electricity using wind power resources and will generate GHG emission reductions by avoiding CO₂ emissions from electricity generation of fossil fuel fired power plants. Operation of this project does not lead to GHG emissions. It is confirmed that the project was not implemented to create GHG emissions primarily for the purpose of its subsequent removal or destruction.

2.2.2 Methodology Deviations

The validation process /3/ has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable large-scale CDM methodology ACM0002 (version 12.1.0) /23/. There was not any methodology deviation applied to this project.

2.2.3 Project Description Deviations

No applicable.

2.2.4 New Project Activity Instances

The project was not a grouped project, hence this clause is not applicable.

2.3 Validation Conclusion

DNV is able to confirm the project is in compliance with the VCS requirements for validation /22/ /25/ /26/.

3 VERIFICATION PROCESS

3.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the registered CDM-PPD, the applied methodology, and the VCS Standard version 3.3 and other relevant VCS requirements.

- A desk review of the monitoring report and all support documents.
- Follow-up interviews with project stakeholders and site inspection.
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records.
- The calibration and maintenance records for the monitoring instruments.
- The management systems to support the project operation and monitoring.

Verification team

Role	Last Name	First Name	Country	Type of involvement					
				Desk review	Site visit	Reporting	Supervision of work	Technical review	TA 1.2 competence
Team leader (Validator & Verifier)	Yang	Xiaoshan	China	✓	✓	✓	✓		✓
Technical reviewer	Li	Lei	China					✓	✓

Duration of verification

Preparations: From 19 December 2012 to 25 March 2013

On-site verification: On 26 March 2013

Reporting, calculation checks and QA/QC: From 27 March 2013 to 23 April 2013

3.2 Document Review

The verification process includes the desk review of the VER/VCU monitoring report (version 1.0 dated 19 December 2012 and updated version 02 dated 3 April 2013) /1/, emission reduction calculation spreadsheet /5/ and other supporting documents and data /15/-/30/. Further, on-site assessments and interviews with those involved in project management and operation were conducted /31/ /32/. This follows summarization of desk review and findings of on-site assessment (i.e. CARs, CLs, and FARs, refer to the Appendix A of this report). Upon successful closing of those findings raised, the final verification report is prepared based on the updated VER/VCU MR (version 02 dated 3 April 2013).

The following documents were assessed as a part of the verification audit:

- The VER/VCU Monitoring Report of version 1.0 dated 19 December 2012 and updated version 02 dated 3 April 2013 for the period from 20 January 2010 to 30 January 2012 /1/;
- Original daily and monthly reading records, and sales records of electricity exported and imported by the project activity during the period from 20 January 2010 to 30 January 2012 /15/ /16/.
- Emission reduction calculation spreadsheet for Heilongjiang Wangyufeng Wind Power Project /5/, completed by Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.. Emission reductions are calculated from the net amount of electricity supplied to the grid multiplied by the ex-ante validated emission factor of the NECPG determined during the CDM validation stage /3/ /24/.

In addition, the project's CDM Project Design Document (PDD) (version 02 dated 1 August 2011) /2/, in particular the monitoring plan included in the CDM PDD, and the CDM validation report (revision 01 dated 4 January 2012) /3/ were also assessed.

3.3 Interviews

On 26 March 2013, DNV performed the on-site visit of the project at Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province of China and conducted an interview with the representatives of the project owner and the CDM consultant. The personnel interviewed during the audit are listed in the reference section of this report /31/, /32/. Main topics of the interview cover implementation of the project construction, applicability of selected methodology, implementation of project monitoring, emission reduction calculation, etc.

3.4 Site Inspections

The on-site assessment involved:

- Assessment of the implementation and operation of the CDM project activity as per the registered CDM-PDD;
- Review of information flows for generating, aggregating and reporting the monitoring parameters;
- Interviews with relevant personnel to confirm that the operational and data collection procedures are implemented in accordance with the monitoring plan in the CDM-PDD;
- A cross-check between information provided in the monitoring report and data from other sources such as plant log books, inventories, purchase records or similar data sources;
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the CDM-PDD and the selected methodology;
- Review of calculations and assumptions made in determining the GHG data and emission reductions;
- Identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

The analysis of documentation, interviews and site visit allowed the assessment of the following processes and assumptions (including QA/QC related issues):

Electricity exported to and imported from the grid was verified to be continuously monitored. The quantity of electricity exported to the grid and imported from the grid during this monitoring period have been cross-checked against the sales receipts /16/. The net electricity supplied to the grid is the difference between the electricity exported to and imported from the grid by the project activity. The emission reductions generated by the project activity during this monitoring period are equal to the amount of net electricity supplied to the grid multiplied by the grid emission factor (1.0280 tCO₂e/MWh) determined ex-ante in the registered PDD /2/ /9/.

3.5 Resolution of Any Material Discrepancy

A corrective action request (CAR) is issued, where:

- i. Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- ii. Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;
- iii. Issues identified in a FAR during validation to be verified during verification have not been resolved by the project participants.

A clarification request (CL) shall be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

One CAR and seven CLs were raised in this verification under the Verified Carbon Standard (VCS) version 3.3 and have been adequately addressed by the project participants by revising the VER/VCU monitoring report (refer to Appendix A) and provide further supporting evidences. No forward action request (FAR) was issued for the pre-CDM project. DNV was able to verify that the GHG emission reductions reported for the “Heilongjiang Wangyufeng Wind Power Project” in the VER/VCU monitoring report 20 January 2010 to 30 January 2012 are fairly stated /1/.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

As part of the site visit DNV was able to confirm that the project implementation is in accordance with the project description contained in registered CDM-PDD of 1 August 2011, and that the monitoring has been carried out in accordance with the monitoring plan contained in the registered CDM-PDD of 1 August 2011 /2/.

The project is a wind farm plant developed by Hegang Longyuan Wind Power Co., Ltd.. The total installed capacity of the project is 49.3 MW, consisting of 58 sets of turbines with rated capacity of 0.85 MW each (type: 46 sets of G52-850 and 12 sets of G58-850) /11/ /19/. The details of the turbines with respect to their numbers, type and capacity have been verified to be consistent with the details provided in the registered CDM-PDD /2/ through the physical observation at project site and reviewing the turbine purchase agreement /11/.

The project intends to reduce GHG emissions by generating electricity from a newly built wind power project to supply the electricity to the NECPG through the on-site 110 kV switchgear. The electricity generated is supplied to the NECPG (which is dominated by coal-fired power plants) under the Power Purchase Agreement signed between the project owner and Heilongjiang Province Electric Power Co., Ltd. /6/. The electricity with voltage level of 690 V generated by the wind turbines is boosted to 35 kV through the on-site 35 kV box transformers equipped with for each turbine; the electricity with voltage level of 35 kV is boosted to 110 kV by the on-site 110 kV transformer.

While the project activity is the installation of a newly grid-connected wind power plant, the project reduces GHG emissions by displacing equivalent electricity that would have been generated in the existing fossil fuel fired power plants or by the addition of new generation in the grid. The project's emission reductions are determined by multiplying the quantity of net electricity supplied by the project ($EG_{\text{facility},y}$) with an ex-ante fixed grid emission factor ($EF_{\text{grid,CM},y}$) of 1.0280 tCO₂e/MWh.

The CDM baseline and monitoring methodology applied is ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” (version 12.1.0) /23/.

The monitoring procedures of electricity supplied to the grid were verified during the site visit and considered adequate. The calibrations of the electricity meters, M1 and M2 installed at the project site (as illustrated in the registered CDM-PDD which was verified on site visit), were carried out and assured by Hegang Metrological Verification Institution /21/. The qualification of the entity performing the calibrations was verified against the accreditation certificate from Heilongjiang Bureau of Quality and Technical Supervision /20/.

The control system at the wind power plant is automated and assures continuous operation due to the availability of wind resource.

4.1.1 Project Implementation in accordance with the registered project design document

DNV verified that the project is fully implemented according to the description in the registered CDM-PDD /2/. The verification team confirmed through visual inspection and document review that all physical features of the CDM project activity including data collection systems and storage systems have been implemented in accordance with the registered CDM-PDD. The project is completely operational which was confirmed by means of the on-site visit. Neither a notification nor approval of change has been requested to the CDM Executive Board.

The emissions reductions reported in this monitoring period are 195 043 tonnes of CO₂ equivalents in the period 20 January 2010 to 30 January 2012 (i.e. 741 days). The yearly expected emission reductions in the registered PDD are 123 430 tonnes of CO₂ equivalents, which corresponds to the emission reductions of 250 580 tonnes of CO₂ equivalents in 741 days, and hence the reported emission reductions are lower than expected. The decrease of the emission reductions in this monitoring period was assessed by DNV in the following section 4.4.

4.1.2 Compliance of monitoring plan with monitoring methodology

DNV is able to confirm that the monitoring plan contained in the registered CDM-PDD of 1 August 2011 /2/ is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 12.1.0) /23/. Neither a revision nor a deviation to the monitoring plan has been requested to CDM Executive Board.

According to the monitoring plan of the registered CDM-PDD /2/, there is three parameters to be monitored: EG_{facility,y}, the quantity of net electricity generation supplied by the project plant/unit to the grid in year y;

EG_{export,y}, electricity exported to the grid by the project;

EG_{import,y}, electricity utilized by the project.

The following table is related to the parameter stipulated in the monitoring plan / methodology:

	Assessment/ Observation
Data / Parameter: (as in monitoring plan):	EG _{facility,y} : quantity of net electricity generation supplied by the project plant/unit to the grid NECPG in year y; EG _{export,y} , electricity exported to the grid by the project;

	$EG_{import,y}$, electricity utilized by the project. $(EG_{facility,y} = EG_{export,y} - EG_{import,y})$
Measuring frequency:	Continuously measured
Reporting frequency:	Monthly recorded and reported
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes
Type of monitoring equipment:	Bi-directional electricity meters /7/
Is accuracy of the monitoring equipment equal or more accurate as stated in monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practise?	The accuracy level of meters (M1 and M2) is 0.5S, which is in line with the accuracy degree not less than 0.5% specified in the monitoring plan of the registered CDM-PDD /2/ .
Calibration frequency /interval:	Once a year
Is the calibration interval in line with the monitoring plan of the monitoring plan? If the monitoring plan does not specify the frequency of calibration, does the selected frequency represent good monitoring practise?	The calibration interval once a year was in line with the monitoring plan of the registered CDM-PDD /2/.
Company performing the calibration:	The certificate of Metrological Authorization for calibrator (Hegang Metrological Verification Institution) was accredited by Heilongjiang Bureau of Quality and Technical Supervision on 31 March 2009 (with validity up to 30 March 2012) /20/, which has been verified by DNV.
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes.
Is (are) calibration(s) valid for the whole reporting period? If the calibration does not cover the whole period please indicated how the project has revised its GHG calculations for the period that calibration was not valid.	Yes. After verifying the calibration records /21/, DNV was able to confirm that the calibrations are valid for the whole reporting period from 20 January 2010 to 30 January 2012.
If applicable, has the reported data been cross-checked with other available data?	Yes. The reported data has been cross-checked with sales receipts. Exported electricity of 190 390.860 MWh and imported electricity of 659 340

	MWh during this monitoring period are indicated in the sales receipts. These values are in line with the data reported in the MR.
How were the values in the monitoring report verified?	As per the registered CDM-PDD /2/, the exported electricity and imported electricity are measured by the meters installed at the project site. The net electricity is determined by the electricity exported to the grid minus the electricity imported from the grid, which is the basis for the calculation of the emission reductions. The input data for the electricity exported and imported by the project in the ER calculation spreadsheet /5/ were taken from original daily and monthly monitoring records /15/, which were further cross-checked against the sales receipts for monthly electricity exported and imported by the project /16/. DNV verified that the values of the exported and imported electricity during this monitoring period reported in the monitoring report are consistent with those data sources /5/ /15/ /16/. All original daily and monthly monitoring records /15/, sales receipts for monthly electricity exported and imported by the project /16/ have been verified by DNV on the project site during the verification site visit. DNV was able to confirm the calculating process and the conclusion is transparent and correct.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	In order to monitor the net electricity supplied by the project to the NECPG, Two bi-directional electricity meters are installed at the on-site 110 kV substation (the main meter M1 with accuracy of 0.5S installed at outlet of the main transformer and the meter M2 with accuracy of 0.5S installed at the inlet of the main transformer) to continuously measure electricity exported to and imported from the grid by the project activity, which is verified to be in accordance with the registered PDD. The meters were continuously monitored and monthly recorded by the staff on duty. On cut-off time at 24:00 of the last day of every month, the project owner and the grid company jointly read the meter M1. After the readings were confirmed by both sides the grid company issued the sales receipts for monthly electricity exported to

	the grid and imported from the grid by the project /16/. During the site visit, the original daily and monthly monitoring records during the monitoring period /15/ have been cross-checked by DNV with sales receipts for monthly electricity exported and imported by the project /16/. DNV was able to confirm that the values of electricity exports and imports are consistent between the original daily and monthly monitoring records /15/ and the sales receipts for monthly electricity exported to the grid and imported from the grid by the project /16/. The meters are jointly inspected and sealed on behalf of the parties and are not interfered with by either party except in the presence of the other party or its accredited representatives. All relevant documents and records are stored electronically and will be kept for two years after the end of the CDM crediting period. The management system for the CDM project was in place, which has been verified during the site visit. In order to make the monitoring system more efficient, before operation of this project, the project owner introduced the CDM Monitoring and management procedure /13/. In the procedure, the organization structure with the responsibilities, personnel competencies, monitoring procedure, calculation procedure and QA/QC procedure have been properly identified By interviewing with some staff /31/ /32/ on site and checking of VER/VCU monitoring reports /1/, training records /12/ and CDM monitoring and management procedure /13/, DNV was able to confirm that the monitoring management system is implemented in compliance with the registered CDM PDD.
In case only partial data are available because activity levels or non-activity parameters have not been monitored in accordance with the registered monitoring plan, has the most conservative assumption theoretically possible been applied or has a request for deviation been approved?	Not applicable. A complete set of data is available in this monitoring period.

4.2 Accuracy of GHG Emission Reduction or Removal Calculations

DNV assessed the different areas that can affect the final emission reduction calculations. All the possible factors were assessed thoroughly to check the accuracy of calculations:

- a) *Electricity generation readings:* The bidirectional electricity meters, which measure the electricity supplied to and imported from the grid by the project, are installed at the on-site substation. Calibration reports of the installed meters have been verified by DNV on site visit.
- b) *Date Recording:* The original meter readings were recorded at 24:00 on the last day of every month.
- c) *Grid emission factor:* The emission factor is determined *ex-ante* as 1.0280 tCO₂e/MWh in the registered CDM PDD /2/ /9/.

Based on the above discussion, it is DNV's opinion that for the current verification period (20 January 2010 to 30 January 2012), the project developer has taken sufficient steps to ensure the accuracy of reported emission reductions.

4.3 Quality of Evidence to Determine GHG Emission Reductions or Removals

The emission reductions for this project are determined by the grid emission factor multiplied by the net electricity delivered to the grid. The emission factor was fixed for the first 7 years renewable CDM crediting period from 31 January 2012 to 30 January 2019 as per the registered PDD /2/. The emission factor fixed *ex-ante* is also applicable to this monitoring period, therefore, the only parameter to be monitored for this project is the net electricity delivered to the grid.

The emission reductions ER_y by the project activity during the crediting period are the difference between baseline emissions (BE_y), project emissions (PE_y) and emissions due to leakage (L_y), as follows:

- a) *Baseline emissions:* Baseline emissions (BE_y in t CO₂e) are the product of the baseline emission factor ($EF_{grid,CM,y}$ in tCO₂e/MWh) times the net electricity supplied by the project activity to the grid ($EG_{facility,y}$ in MWh, calculated as difference of electricity exported to and imported from the grid).
- b) *Project emissions:* The project emissions are regarded as zero as per the applied methodology since the project is a greenfield wind power project.
- c) *Leakage:* No leakage has to be considered for the project activity as per the applied methodology.
- d) *Emission reductions:* $ER_y = BE_y - PE_y - L_y = BE_y$.

The combined margin of 1.0280 tCO₂e/MWh is calculated according to the baseline methodology ACM0002 (version 12.1.0) /23/.

The total exported electricity to the grid ($EG_{export,y}$) has been verified to be 190 390.860 MWh and the total imported electricity from the grid ($EG_{import,y}$) by the project has been verified to be 659 340 MWh in this monitoring period. The net electricity supplied to the grid ($EG_{facility,y}$) by the project is calculated to be 189 731.520 MWh in this monitoring period. Therefore, the emission reductions during the monitoring period are calculated as:

$$ER_y = BE_y = EF_{grid,CM,y} * EG_{facility,y} = 1.0280 \text{ tCO}_2\text{e/MWh} * 189\,731.520 \text{ MWh} = 195\,043 \text{ tCO}_2\text{e}$$

Thus the claimed emission reductions of 195 043 tCO₂e during the reported monitoring period was verified by reviewing the presented relevant documents /15/ /16/ /17/.

The net electricity supplied to the grid and emission reductions claimed for the monitoring period were as shown in the following table:

Period	Exported electricity to the grid $EG_{\text{export},y}$ (MWh)			Imported electricity from the grid $EG_{\text{import},y}$ (MWh)		
	Values from sales receipts /16/	Values from monthly records /15/	Verified	Values from sales receipts /16/	Values from monthly records /15/	Verified
	A	B	$C=\text{Min}(A, B)$	D	E	$F=\text{Max}(D, E)$
20/01/2010-31/01/2010	1 309.440	1 309.440	1 309.440	4.620	4.620	4.620
01/02/2010-28/02/2010	3 784.440	3 784.440	3 784.440	50.160	50.160	50.160
01/03/2010-31/03/2010	8 649.960	8 649.960	8 649.960	35.640	35.640	35.640
01/04/2010-30/04/2010	8 914.620	8 914.620	8 914.620	11.220	11.220	11.220
01/05/2010-31/05/2010	7 288.380	7 288.380	7 288.380	14.520	14.520	14.520
01/06/2010-30/06/2010	4 492.620	4 492.620	4 492.620	31.680	31.680	31.680
01/07/2010-31/07/2010	3 746.820	3 746.820	3 746.820	48.180	48.180	48.180
01/08/2010-31/08/2010	5 790.840	5 790.840	5 790.840	15.840	15.840	15.840
01/09/2010-30/09/2010	8 755.560	8 755.560	8 755.560	21.780	21.780	21.780
01/10/2010-31/10/2010	9 327.120	9 327.120	9 327.120	15.180	15.180	15.180
01/11/2010-30/11/2010	10 722.360	10 722.360	10 722.360	31.680	31.680	31.680
01/12/2010-31/12/2010	9 109.980	9 109.980	9 109.980	62.700	62.700	62.700
01/01/2011-31/01/2011	13 377.540	13 377.540	13 377.540	19.800	19.800	19.800
01/02/2011-28/02/2011	8 490.900	8 490.900	8 490.900	26.400	26.400	26.400
01/03/2011-31/03/2011	9 119.220	9 119.220	9 119.220	11.220	11.220	11.220
01/04/2011-30/04/2011	7 305.540	7 305.540	7 305.540	25.740	25.740	25.740
01/05/2011-31/05/2011	6 717.480	6 717.480	6 717.480	20.460	20.460	20.460
01/06/2011-30/06/2011	4 996.200	4 996.200	4 996.200	24.420	24.420	24.420
01/07/2011-31/07/2011	2 558.160	2 558.160	2 558.160	35.640	35.640	35.640
01/08/2011-31/08/2011	4 386.360	4 386.360	4 386.360	43.560	43.560	43.560
01/09/2011-30/09/2011	8 696.160	8 696.160	8 696.160	11.220	11.220	11.220
01/10/2011-31/10/2011	10 416.780	10 416.780	10 416.780	8.580	8.580	8.580

01/11/2011-30/11/2011	10 862.940	10 862.940	10 862.940	13.860	13.860	13.860
01/12/2011-31/12/2011	10 649.100	10 649.100	10 649.100	31.680	31.680	31.680
01/01/2012-30/01/2012	10 922.340	10 922.340	10 922.340	43.560	43.560	43.560
<i>Total</i>			190 390.860			659 340

The emissions reductions reported in this monitoring period (i.e. 741 days) are 195 043 tonnes of CO₂e. The yearly expected emission reductions in the registered PDD are 123 430 tonnes of CO₂ equivalents, which corresponds to the emission reductions of 250 580 tonnes of CO₂ equivalents in 741 days, and hence the reported emission reductions are lower than expected. The lower power generation in the monitoring period was mainly due to lower wind resources in the project area as indicated by the wind data from local meteorological bureau /18/. DNV was able to confirm that the power generation and emission reduction achieved by the project activity in this monitoring period was reasonable.

Years	GHG emission reductions or removals (tCO ₂ e) verified
Year 2010 (20/01/2010-31/12/2010, i.e. 346 days)	83 832
Year 2011 (01/01/2011-31/12/2011, i.e. 365 days)	100 028
Year 2012 (01/01/2012-30/01/2012, i.e. 30 days)	11 183
Total estimated ERs (in 741 days)	195 043

4.4 Management and Operational System

The management system for the project is in place. In order to make the monitoring system more efficient, before operation of this project, the project owner edited the CDM Monitoring and management procedure /13/. In this procedure, the organization structure with the responsibilities, personnel competencies, monitoring procedure and monitoring management have been properly identified. By interviewing with the staff and checking during the site visit, DNV can confirm that the monitoring management system is implemented following the CDM Monitoring and management procedure /13/.

Monitoring and reporting of electricity generation is part of normal operations of the project. The quality of meter readings is assured through calibration of electricity meters /21/ and through cross checking of readings between the meter and the sales receipts /16/.

5 VERIFICATION CONCLUSION

DNV Climate Change Services AS (DNV) has performed the verification of the emission reductions that have been reported for the “Heilongjiang Wangyunfeng Wind Power Project” (CDM Registration No. 5711) for the period 20 January 2010 to 30 January 2012.

In addition, DNV has also carried out a validation of the project with regard to VCS requirements that apply to projects already registered under the CDM. DNV was able to confirm that the project is in compliance with the VCS requirements for validation of projects already registered under the CDM.

The project participants are responsible for the collection of data in accordance with the monitoring plan and the reporting of GHG emissions reductions from the project.

It is DNV’s responsibility to express an independent verification statement on the reported GHG emission reductions from the project.

DNV conducted the verification on the basis of the approved methodology ACM0002 (version 12.1.0), the monitoring plan contained in the registered CDM-PDD of 1 August 2011 and the monitoring report (version 02) dated 3 April 2013. The verification included i) checking whether the provisions of the monitoring methodology and the monitoring plan were consistently and appropriately applied and ii) the collection of evidence supporting the reported data.

DNV’s verification approach draws on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. DNV planned and performed the verification by obtaining evidence and other information and explanations that DNV considers necessary to give reasonable assurance that reported GHG emission reductions are fairly stated.

In our opinion the GHG emissions reductions of the “Heilongjiang Wangyunfeng Wind Power Project” for the period 20 January 2010 to 30 January 2012 are fairly stated in the monitoring report (version 02) dated 3 April 2013.

The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 12.1.0) and the monitoring plan contained in the registered CDM-PDD of 1 August 2011.

DNV Climate Change Services AS is able to certify that the emission reductions from the “Heilongjiang Wangyunfeng Wind Power Project” during the period 20 January 2010 to 30 January 2012 amount to 195 043 tonnes of CO₂ equivalent, detailed as below.

Reporting period: 20 January 2010 to 30 January 2012

Verified GHG emission reductions or removals in the above reporting period:

Years	GHG emission reductions or removals (tCO ₂ e) verified
Year 2010 (20/01/2010-31/12/2010, i.e. 346 days)	83 832
Year 2011 (01/01/2011-31/12/2011, i.e. 365 days)	100 028
Year 2012 (01/01/2012-30/01/2012, i.e. 30 days)	11 183
Total estimated ERs (in 741 days)	195 043

DNV does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to DNV and the engagement conditions detailed in this report. DNV cannot be held liable by any party for decisions made or not made based on this report.

6 REFERENCES

Documents provided by the Project Participants that relate directly to the GHG components of the project. These have been used as direct sources of evidence for the periodic verification conclusions, and are usually further checked through interviews with key personnel.

- /1/ Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: VER/VCU Monitoring Report for VCS verification of Heilongjiang Wangyunfeng Wind Power Project, version 1.0 dated 19 December 2012 and version 02 dated 3 April 2013.
- /2/ Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: Registered CDM-PDD for Heilongjiang Wangyunfeng Wind Power Project, version 02 dated 1 August 2011.
- /3/ Bureau Veritas Certification Holding SAS (BVCH): Validation Report of the CDM project, Heilongjiang Wangyunfeng Wind Power Project, validation Report No.: CHINA-Val/0231/2009, revision 01 dated 4 January 2012.
- /4/ UNFCCC interface: CDM project information for Heilongjiang Wangyunfeng Wind Power Project (reference No. 5711). The project was registered on UNFCCC on 31 January 2012 with a renewable crediting period starting from 31 January 2012 to 30 January 2019.
<http://cdm.unfccc.int/Projects/DB/BVQI1327230197.63/view>
- /5/ Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: Emission Reduction calculation spreadsheet for VCS Heilongjiang Wangyunfeng Wind Power Project, version 01 dated 19 December 2012 and version 02 dated 3 April 2013.
- /6/ Hegang Longyuan Wind Power Co., Ltd. and Heilongjiang Province Electric Power Co., Ltd. (on behalf of NECPG): Power Purchase Agreement, dated 17 May 2010.
- /7/ Hegang Longyuan Wind Power Co., Ltd.: clarification of installed monitoring equipment including electricity meters and instrument transformer, dated 7 April 2013
- /8/ NDRC (DNA of China): Letter of approval (LoA) for Heilongjiang Wangyunfeng Wind Power Project (English version Ref. No. 2342), issued in January 2010.
- /9/ NDRC: Baseline emission factor calculation result of China Power Grids, dated 2 July 2009.
- /10/ Hegang Longyuan Wind Power Co., Ltd.: Project construction-start application and permission, dated 4 June 2009.
Zhongrui Yuehua Certified Public Accountants: Verification report of project construction completion of Heilongjiang Wangyunfeng Wind Power Project, dated 31 October 2010.
- /11/ Hegang Longyuan Wind Power Co., Ltd. and Gamesa Wind (Tianjin) Co., Ltd.: Technical agreement of wind turbines purchase contract for the project activity Heilongjiang Wangyunfeng Wind Power Project, dated 12 May 2009.
- /12/ Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: Dedicated plan for regular CDM trainings related to the project monitoring, dated 18 April 2009; Training records for the project monitoring, in December 2009
- /13/ Hegang Longyuan Wind Power Co., Ltd.: CDM Monitoring and Management Procedure issued on 10 May 2010 and entered into effect on 1 June 2010.
- /14/ Municipal Administration for Industry and Commerce of Hegang: Business license (Reg. No.: 230000400006771, established on 3 June 2009, with validity period from 3 June 2009 to 2 June 2029.
- /15/ Hegang Longyuan Wind Power Co., Ltd.: Raw daily and monthly reading records of electricity exported to the grid and imported from the grid by the project during January 2010 to January

2012.

- /16/ Hezhang County Power Supply Bureau (It belongs to Heilongjiang Province Electric Power Co., Ltd.): sales receipts for monthly electricity exported and imported by the project during the period of January 2010 to January 2012.
- /17/ Hegang Longyuan Wind Power Co., Ltd.: Operation logs for project activity "Heilongjiang Wangyunfeng Wind Power Project", during the period of January 2010 to January 2012.
- /18/ Meteorological Bureau of Luobei County: wind statistics of Luobei County, dated 25 February 2013.
- /19/ Gamesa Wind (Tianjin) Co., Ltd.: Technical parameter of installed wind turbine for the project.
- /20/ Heilongjiang Bureau of Quality and Technical Supervision: Accreditation certificate of Metrological Authorization to The Legal Metrological Verification Institution (Hegang Metrological Verification Institution), Hei FaJi 2009 No. D010, with validity period from 31 March 2009 to 30 March 2012, issued on 31 March 2009.
- /21/ Hegang Metrological Verification Institution: Electric power meters calibration reports,
 Calibration Certificate (Certificate No. 090515000085) for meter M1 (Type DSSD719, Serial No. 4200525040909208867, accuracy 0.5S) calibrated on 20 May 2009 with validity up to 19 May 2010;
 Calibration Certificate (Certificate No. 090515000025) for meter M2 (Type DSSD719, Serial No. 4200222040909208865, accuracy 0.5S) calibrated on 20 May 2009 with validity up to 19 May 2010;

 Calibration Certificate (Certificate No. 100515000088) for meter M1 (Type DSSD719, Serial No. 4200525040909208867, accuracy 0.5S) calibrated on 17 May 2010 with validity up to 16 May 2011;
 Calibration Certificate (Certificate No. 100515000022) for meter M2 (Type DSSD719, Serial No. 4200222040909208865, accuracy 0.5S) calibrated on 17 May 2010 with validity up to 16 May 2011;

 Calibration Certificate (Certificate No. 110515000008) for meter M1 (Type DSSD719, Serial No. 4200525040909208867, accuracy 0.5S) calibrated on 15 May 2011 with validity up to 14 May 2012;
 Calibration Certificate (Certificate No. 110515000002) for meter M2 (Type DSSD719, Serial No. 4200222040909208865, accuracy 0.5S) calibrated on 15 May 2011 with validity up to 14 May 2012;

 Calibration Certificate (Certificate No. 120818105702) for meter M1 (Type DSSD719, Serial No. 4200525040909208867, accuracy 0.5S) calibrated on 12 May 2012 with validity up to 11 May 2013;
 Calibration Certificate (Certificate No. 120818100006) for meter M2 (Type DSSD719, Serial No. 4200222040909208865, accuracy 0.5S) calibrated on 12 May 2012 with validity up to 11 May 2013.

Background documents related to the design and/or methodologies employed in the design or other reference documents.

- /22/ CDM Executive Board: CDM Validation and Verification Standard, version 03.0, CDM-EB65-A04-STAN.
- /23/ CDM Executive Board: "Consolidated baseline methodology for grid-connected electricity generation from renewable sources" ACM0002, version 12.1.0, dated 26 November 2010.
- /24/ CDM Executive Board: Tool to calculate the emission factor for an electricity system, version 02.2.1.
- /25/ Verified Carbon Standard – VCS Standard: VCS version 3.3, dated 4 October 2012.
- /26/ Verified Carbon Standard – VCS Program Guide: VCS version 3.4, dated 4 October 2012.
- /27/ Verified Carbon Standard – VCS Sectoral Scopes, <http://v-c-s.org/node/448>.
- /28/ Verified Carbon Standard – VCS Monitoring Report Template, version 3.2
- /29/ Verified Carbon Standard – VCS Verification Report Template, version 3.2
- /30/ Verified Carbon Standard – Registration and Issuance Process: VCS version 3.4, 4 October 2012.

Persons interviewed during the initial verification, or persons who contributed with other information that are not included in the documents listed above.

- /31/ Hegang Longyuan Wind Power Co., Ltd.:
 Mr. Wang Zhewei, Project Manager
 Mr. Feng Jialiang, Engineer
- /32/ Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.:
 Mr. Guo Huidong, Project Manager

APPENDIX A

CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Corrective action requests

CAR ID	Corrective action request	Response by Project Participants	DNV's assessment of response by Project Participants
CAR 1	The meter M2 having serial number of 4200525040909208865 was reported to be used by the project in the monitoring report version 01. However, the calibration reports provided and the meter verified on site visit indicate the serial number of 4200222040909208865. Corrective action is requested.	The serial number of meter M2 should be 4200222040909208865. It is revised in version 02 of the monitoring report.	OK. The calibration reports, meter in use and meter readings for meter M2 have been verified on site, thus DNV was able to confirm the typo in the monitoring report version 1.0. The revised monitoring report was corrected on this. CAR 1 is therefore closed.

Clarification requests

CL ID	Clarification request	Response by Project Participants	DNV's assessment of response by Project Participants
CL 1	Clarification request is sought for the qualification certificate of the entity that performed calibrations of electricity meters for the project activity.	The qualification certificate of the entity that performed calibrations of electricity meters for the project activity is submitted as an attachment.	OK. The qualification certificates of the entity (Hegang Metrological Verification Institution) have been verified. Hegang Metrological Verification Institution was accredited for calibration of the electricity meters installed for the project activity. The calibrations were performed within the validity period of the qualification certificates /20/. CL 1 is therefore closed.
CL 2	Clarification is sought for the calibrations of electricity meters for the project activity.	Please see the attached calibrations of electricity meters for year 2009-2012.	OK. The calibration reports for the monitoring

	Calibration reports provided cannot cover the whole monitoring period.		period have been verified. Validity period of the calibrations covers the whole monitoring period. CL 2 is therefore closed.
CL 3	Clarification is sought for lower power generation in the monitoring period.	The emission reduction difference is mainly due to wind resource fluctuation. According to the monthly wind speed data provided by the local weather station, the average wind resource during this monitoring period is lower than the average annual wind speed based on the long term data listed in the FSR of the project.	OK. The lower power generation in the monitoring period was mainly due to lower wind resources in the project area as indicated by the data from local meteorological bureau /18/. CL 3 is therefore closed.
CL 4	Clarification is sought for individual readings of the electricity meter in the monitoring period to ensure transparency.	Please check the revised ER calculation sheet.	OK. The revised spreadsheet has been verified. The readings were verified to be in line with the original records. CL 4 is therefore closed.
CL 5	Clarification is sought for specifications of equipment installed which was not reported in the monitoring report version 01.	The detailed information about the equipment installed by the project has been added in the updated monitoring report version 02.	OK. The revised monitoring report has been verified. The technical parameters of installed equipment were verified to be consistent with the registered CDM-PDD. CL 5 is therefore closed.
CL 6	It was reported in the monitoring report version 01 that the first wind turbine of the project started commissioning on 28 December 2009, and the project start date was thus identified as the same day. However, the project crediting period and the monitoring period were from 1 January 2010 to 30 January 2012. Clarification is sought for the project start	According to VCS standard 3.3, "the project start date is the date on which the project began generating GHG emission reductions or removals". For the project, the project start date is 20 January 2010 on which the project started generating emission reductions.	OK. It was verified that the project begun generating electricity on 20 January 2010 through raw daily records /15/, as per the VCS standard 3.3, the project start date is determined as 20 January 2010. The updated monitoring report and ER calculation sheet have been verified. CL 6 is therefore closed.

	date.		
CL 7	Clarification is sought for forms of GHG related environmental credits of the project.	The project's credits under this report apply for VCS standard Credits only	OK. It was also declared in the revised monitoring report. CL 7 is therefore closed.

Forward action requests from previous verification

FAR ID	Forward action request	Summary of how FAR has been addressed in this reporting period	Assessment of how FAR has been addressed
NA	NA	NA	NA

Forward action requests from this verification

FAR ID	Forward action request	Response by Project Participants	DNV's assessment of response by Project Participants
NA	NA	NA	NA